

FPGA2 Week9 Team

2021041091_한규철, 2021041089_최태민

동작영상 유튜브 주소

존슨 카운터 영상 : https://youtu.be/vCX_CYNrquI?si=EhplESpn0wScBbYP

실습 응용 링카운터 영상 : <https://youtu.be/B00bfulwpys?si=VoyKOj2QwJHsFFvj>

실습 응용 리플카운터 영상 : <https://youtu.be/BZC5qBIQfkW?si=xXYTVWeW8OODsSJy>

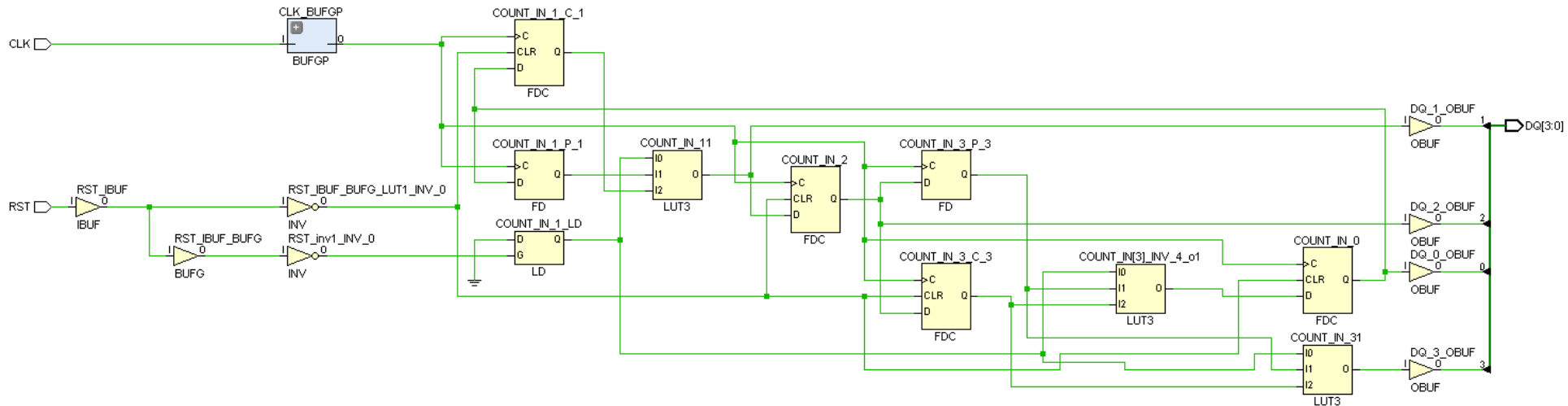
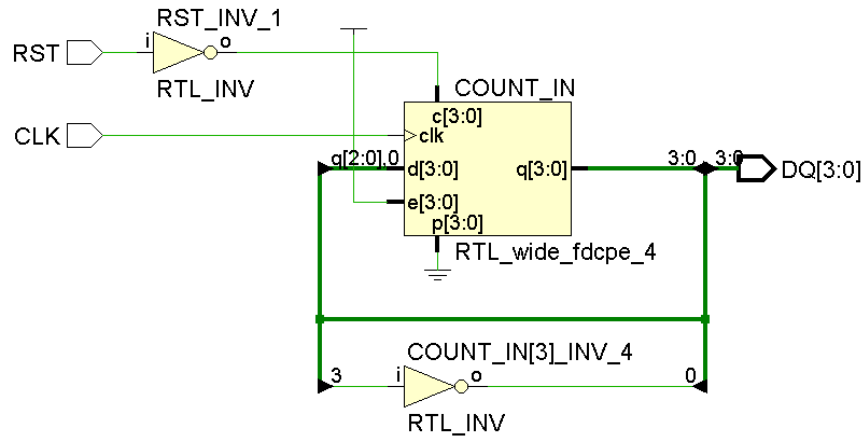
존슨 카운터 동작테이블

입력	Johnson COUNTER			
CLOCK	Q0	Q1	Q2	Q3
↑	0	0	0	0
	1	0	0	0
	1	1	0	0
	1	1	1	0
	1	1	1	1
	0	1	1	1
	0	0	1	1
	0	0	0	1
	0	0	0	0
	1	0	0	0
	1	1	0	0

존슨 카운터 시뮬레이션화면



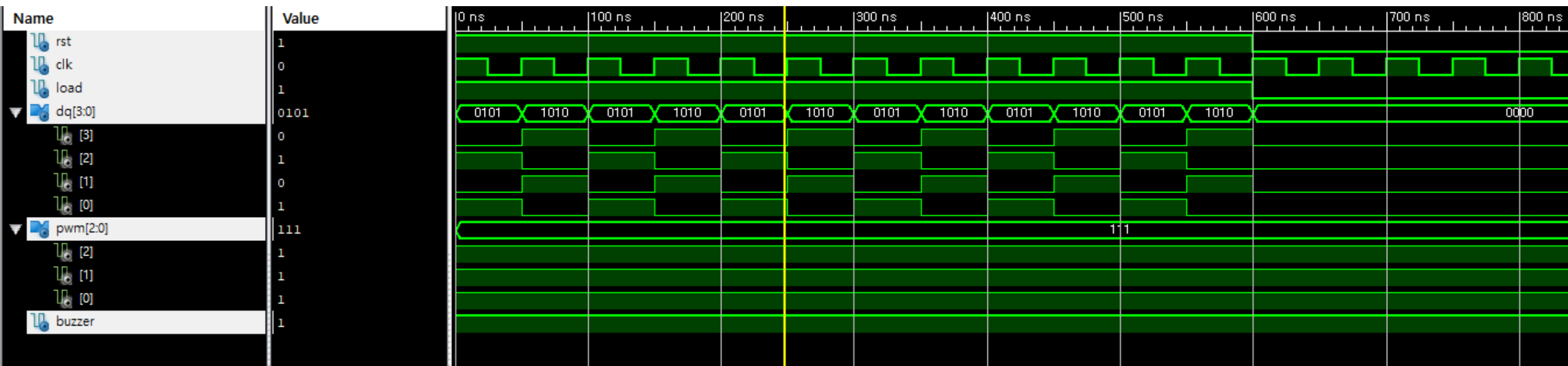
존슨 카운터 회로화면



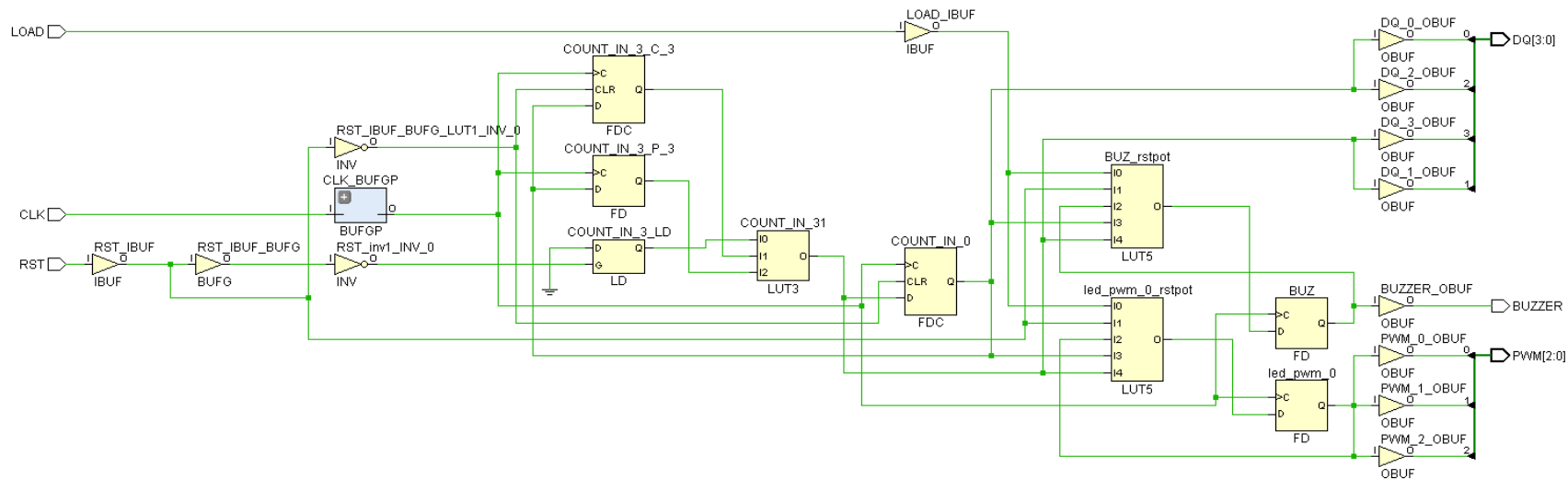
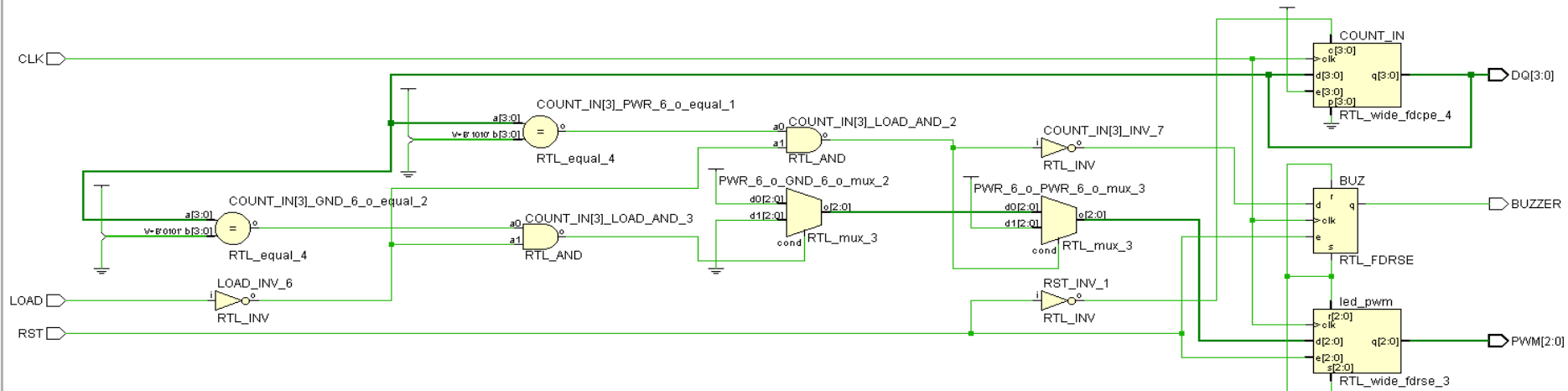
실습응용 링카운터 동작테이블

CLOCK	RESET	LOAD	Y0	Y1	Y2	Y3	BUZ	PWM
0	0	X	0	0	0	0	X	X
0	1	X	X	X	X	X	X	X
1	1	1	1	0	1	0	1	1
1		1	0	1	0	1	1	1
1		1	1	0	1	0	0	1
1		1	0	1	0	1	1	0

실습응용 링카운터 시뮬레이션화면



실습응용 링카운터 회로화면

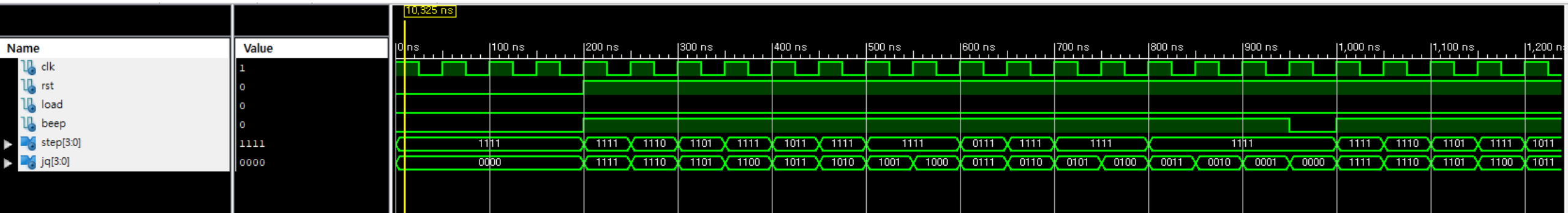


실습응용 리플카운터 동작테이블

입력	Johnson COUNTER							
CLOCK	Q0	Q1	Q2	Q3	Y4	Y5	Y6	Y7
↑	0	1	1	1	0	1	1	1
	1	0	1	1	1	0	1	1
	0	0	1	1	1	1	1	1
	1	1	0	1	1	1	0	1
	0	1	0	1	1	1	1	1
	1	0	0	1	1	1	1	1
	0	0	0	1	1	1	1	1
	1	1	1	0	1	1	1	0
	0	1	1	0	1	1	1	1
	1	0	1	0	1	1	1	1
	0	0	1	0	1	1	1	1
	1	1	0	0	1	1	1	1
	0	1	0	0	1	1	1	1
	1	0	0	0	1	1	1	1
	0	0	0	0	1	1	1	1

입력	Johnson COUNTER					
CLOCK	LOAD	Q0	Q1	Q2	Q3	Beep(부저)
X	1	0	1	1	1	1
	1	1	0	1	1	1
	1	0	0	1	1	1
	1	1	1	0	1	1
	1	0	1	0	1	1
	1	1	0	0	1	1
	1	0	0	0	1	1
	1	1	1	1	0	1
	1	0	1	1	0	1
	1	1	0	1	0	1
	1	0	0	1	0	1
	1	1	1	0	0	1
	1	0	1	0	0	1
	1	1	0	0	0	1
	1	0	0	0	0	1
	1	1	0	0	0	1
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0	0	1	
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1	0	0	0	0	1	
1	0	0	0	0	1	
1	0	0	0			

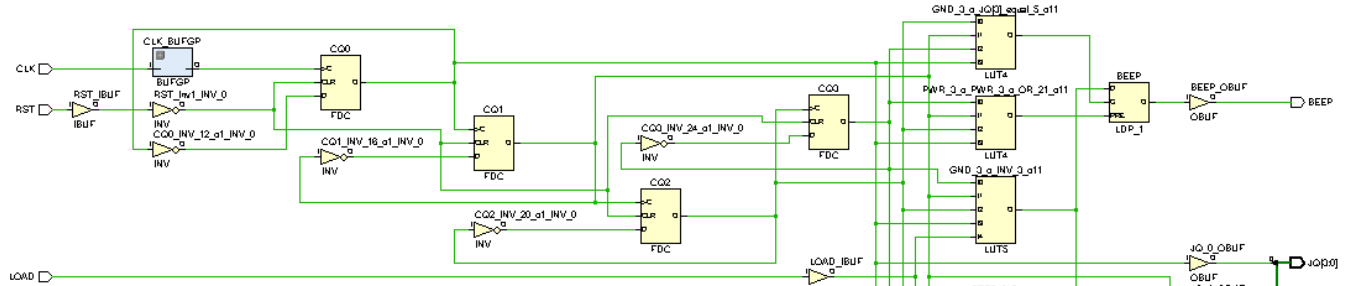
실습응용 리플카운터 시뮬레이션화면



실습응용 리플카운터 회로화면

Schematic x

36 Nets



ice x RTL Schematic x

I/O Ports 55 Nets

